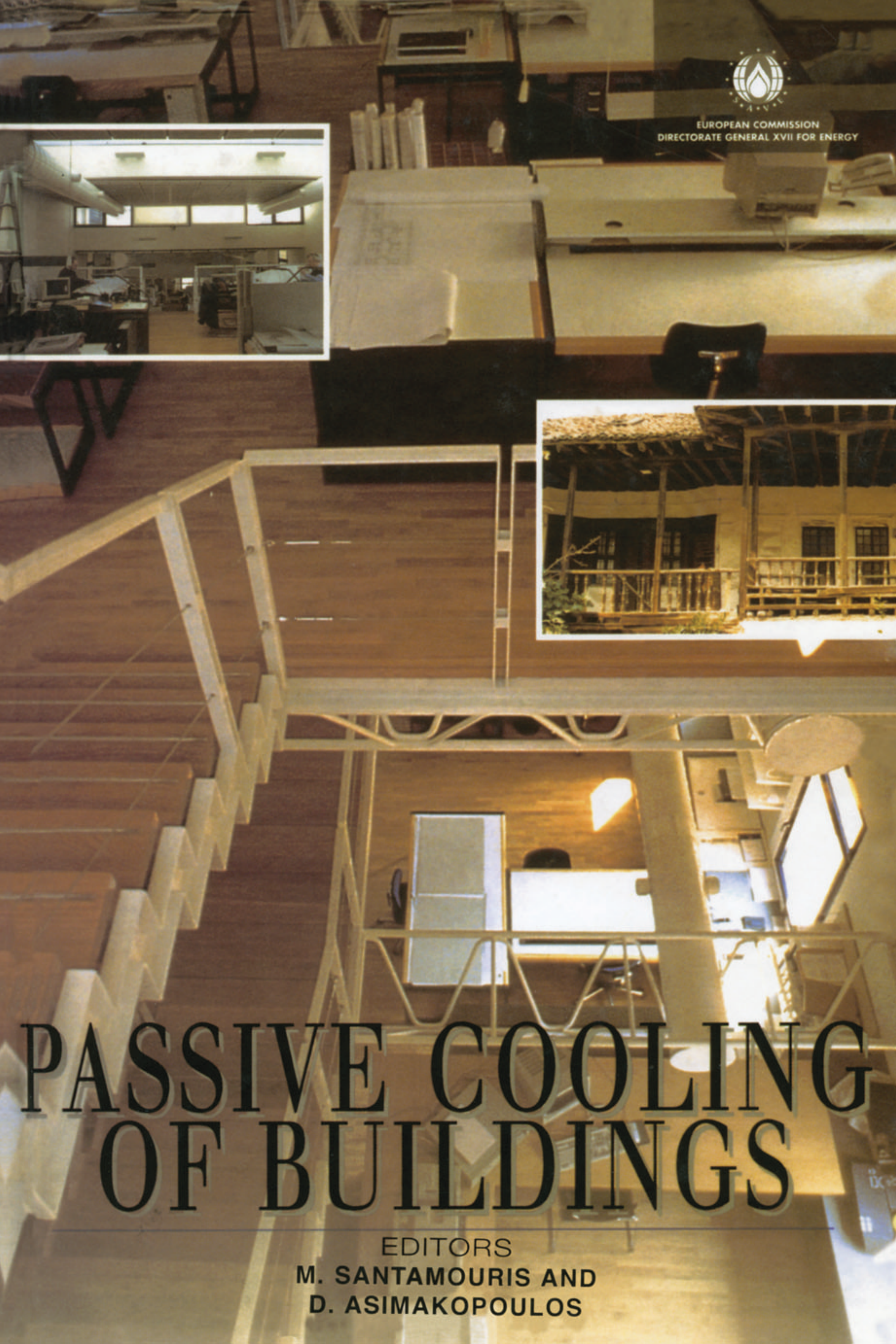

EUROPEAN COMMISSION
DIRECTORATE GENERAL XVII FOR ENERGY

PASSIVE COOLING OF BUILDINGS

EDITORS
M. SANTAMOURIS AND
D. ASIMAKOPOULOS

PASSIVE COOLING OF BUILDINGS

This page intentionally left blank

PASSIVE COOLING OF BUILDINGS

EDITORS
M. SANTAMOURIS AND
D. ASIMAKOPOULOS

earthscan
from Routledge

First published by James & James (Science Publishers) Ltd. in 1996

This edition published 2013 by Earthscan

For a full list of publications please contact:

Earthscan
2 Park Square, Milton Park, Abingdon, Oxon OX14 4RN
Simultaneously published in the USA and Canada by Earthscan
711 Third Avenue, New York, NY 10017

Earthscan is an imprint of the Taylor & Francis Group, an informa business

A catalogue record for this book is available from the British Library

ISBN 978-1-873936-47-4 (hbk)

Typeset by Edgerton Publishing Services, Huddersfield, UK

Table of contents

Preface

The increase in household income in Europe and the relatively low cost of electricity have helped air conditioning to become highly popular. This has resulted in a significant increase in building energy consumption in southern Europe. Sales of air-conditioning equipment in this region have increased considerably over the past few years and are now worth close to 1.7 billion ECU per year. In Greece, for example, while the annual sales of packaged air conditioners was close to 2000 units in 1986, the market jumped to over 100,000 units in 1988.

The specific energy consumption of buildings for cooling needs is dictated by the climate, the type of building and the installed equipment. A comparison of the annual specific energy consumption for cooling in large air-conditioned office buildings in Denmark, Greece, The Netherlands, Norway, Sweden and the UK, shows a variation of between 15 to 110 kWh/m^2. It should be noted that, despite the Mediterranean climate, Greek office buildings do not use more electricity than Swedish or British offices for heating, ventilating and air-conditioning (HVAC) systems during summer. Clearly, the problem of cooling in buildings is not peculiar to southern climates alone. It may be just as important in northern climatic regions, depending on the building type and construction.

The impact of the use of air conditioners on electricity demand is a serious problem for almost all southern European countries. Peak electricity loads force utilities to build additional power plants in order to satisfy the demand, thus increasing the average cost of electricity.

Environmental problems associated with the use of ozone-depleting CFC refrigerants used in conventional air conditioners present an additional argument for minimizing the use of HVAC systems for energy savings in the cooling sector. Problems of indoor air quality associated with the use of air conditioning should also be taken into account. Recent studies of air-conditioned buildings have shown that illness indices are relatively higher for these than those for non-air-conditioned buildings.

Alternative passive cooling techniques, based on improved thermal protection of the building envelope and on the dissipation of building's thermal load to a lower-temperature heat sink, have been proved to be very effective. These strategies and techniques have already reached a certain level of acceptance in architectural and industrial circles. Passive techniques as alternatives to air conditioning can bring important energy, environmental, financial, operational and qualitative benefits.

The study and application of passive cooling is a multilayered and multidisciplinary process. It is important to treat the subject in conjunction with other aspects of architectural design; it should not be considered in isolation. A useful framework for considering passive and hybrid cooling in the context of environmental design can be summarized as follows: prevention of heat gain, modulation of heat gain and heat dissipation. Protection from heat gain may involve:

- landscaping and the use of outdoor and semi-outdoor spaces,
- building form, layout and external finishing,
- solar control and shading of building surfaces,
- thermal insulation, and
- control of internal heat gain.

This book has three main objectives:

- to report on basic knowledge to date in the field of passive cooling, as well as on new tools developed during the past few years;
- to present the recent progress in the field that has been achieved through prominent research programmes; and
- to identify priorities for future research into passive cooling.

The book provides information on all available passive cooling methods and techniques, their potential effectiveness, their basic principles and the criteria needed to identify those most appropriate for specific types of buildings. It also includes presentations of several easy-to-use methods that are available for calculation of the cooling potential of the most important techniques, as well as of the overall thermal performance of buildings.

This book was prepared by the Central Institution for Energy Efficiency Education (CIENE), within the framework of the SAVE Programme 'Short Educational Structure on Energy Efficiency in Buildings' of the European Commission, Directorate General XVII for Energy.

The main goal of the SAVE Programme is to undertake actions that promote the efficient use of energy in all member states of the EU. In this way, the Programme contributes to achieving the aims of the EU energy policy for rational use of natural resources and reduction of carbon dioxide emissions in the atmosphere.

CIENE was established in October 1992 as a SAVE Programme, under the auspices of the National and Kapodistrian University of Athens, Greece. Its operation is harmonized with the philosophy of the SAVE Programme of establishing a network of energy centres throughout Europe. The main role of CIENE is to organize and coordinate specialized training and continuing-education programmes on energy efficiency and energy management.

This book is addressed to anyone who is interested in energy conservation in buildings. The information provided can be used by building owners, occupants and those professionals directly or indirectly involved in the construction, operation and maintenance of buildings. In some respects, this book may be of more interest to

those who have some background experience of energy-related topics and building applications.

We thank the various authors, who devoted a great deal of time to perfecting their chapters. We wish to acknowledge the help, encouragement and support of the DGXVII staff who work for the SAVE Programme, and especially that of Mr E. Dalamangas who has closely followed the overall project.

We would like to acknowledge the help of a group of individuals who have provided valuable contributions, comments and suggestions for this book : S. Alvarez of AICIA Seville, Spain; F. Allard of the University of La Rochelle, France; J. Goulding of University College Dublin, Ireland; M. Grosso of the Politecnico di Torino, Italy; G. Guaraccino of LASH/ENTPE, Lyon, France; O. Lewis of University College Dublin; E. Maldonado of the University of Porto; C.A. Roulet of EPF Lausanne, Switzerland; A. Tombazis of Meletitiki Ltd; P. Wouters and L. Vandaele at BBRI; and S. Yannas of the Architectural Association of London.

Any comments with regard to the contents and structure of this book are welcome.

M. I. Santamouris
D. N. Asimakopoulos
Editors

Athens, January 1996

This page intentionally left blank

1

Cooling in buildings

Ever since humans have moved into shelters, in search of a more stable environment, they have looked for ways to improve indoor conditions. Inevitably, however, the indoor environment is influenced by prevailing outdoor conditions, daily and seasonal changes in climate and varying occupant requirements due to the type and operation of the building. Depending on the location and season, emphasis is given to either cooling or heating of indoor spaces, in an attempt to counterbalance the unfavourable outdoor conditions and achieve indoor comfort by controlling the indoor temperature, humidity, light availability and air quality.

Historically, for practical reasons and owing to natural laws, humans have been most successful in controlling their environment in situations requiring heat. Maintaining a warm environment has always been considered necessary during the cold season. However, in modern society, maintaining a cool environment during the warm months has proven to be as important for the optimum utilization of human resources and productivity.

The greatest advance has been in the change from rather artful design practices of indoor spaces to the detailed analytical methods that are necessary in order to handle complex building structures. Modern buildings, taking full advantage of currently available state-of-the-art technology, provide an indoor environment with high living and working standards.

Air-conditioning (A/C) systems can be used for year-round environmental control, in terms of temperature, moisture content, and air quality. However, like all mechanical systems, A/C systems consume valuable electrical energy for their operation. The shortage of conventional energy sources and escalating energy costs have caused reexamination of the general design practices and applications of A/C systems and the development of new technologies and processes for achieving comfort conditions in buildings by natural means. Indoor thermal comfort implies that humans are satisfied with the prevailing air conditions in the space – neither too warm nor too cool.

The historical development of the various processes and systems for mechanical or passive cooling, along with current trends and practices in the field, are reviewed in the following sections.

HISTORICAL DEVELOPMENT

Cooling is the transfer of energy from the space or the air supplied to the space, in order to achieve a lower temperature and/or humidity level than those of the natural surroundings. The development of cooling processes has passed through several stages, starting from simple intuitive applications of natural cooling techniques, such as shading, evaporative cooling and air circulation for enhancing the comfort sensation, to mechanical cooling systems, known as air conditioners, based on mechanical refrigeration cycles.

In fact, it appears that there has been a return to the utilization of several well-known techniques and processes that were used successfully even in the early periods of civilization. The principles of passive cooling are the same, but they are now enhanced with the available technological know-how and they are optimized so that they can be successfully incorporated into the building design and operation, in a suitable form for providing the best results.

In the early stages of history earth shelters were used by humans as a readily and naturally available living space, which provided protection from high and low temperatures, as well as from other unfavourable weather conditions. Building architecture quickly developed as an art, as the needs and demands of humans were changing with time, along with the appropriate know-how and availability of tools.

Well before the development of mechanical systems, though, several techniques for providing cooling and comfortable indoor conditions were applied in building architecture. The use of these techniques was not based on the understanding of the physical processes involved, but rather on conceptual experience. The majority were simple applications, like air movement through open spaces, external and internal shading, appropriate arrangement of the immediate surrounding spaces (vegetation, open pools and ponds) and use of proper building materials ('cold' marble and light surface colours). In addition, human mobility in indoor spaces was used extensively in order to avoid spaces with uncomfortable conditions during the day, as a result of direct solar gains.

Building design incorporated various fundamental and simple, but effective, principles. The large openings of the buildings, allowed for ample cross air movement, which can have a significant cooling effect. Even if the outdoor air is not at the desired temperature, air movement creates a cooling sensation as it moves around the human body.

The building itself provided sufficient protection to the occupants by properly shading the living spaces from direct solar gains. The landscaping around the buildings was primarily designed for aesthetic reasons, but at the same time improved the microclimate around the building, by providing shading and evaporative cooling. Extensive use of vegetation around the buildings provided the necessary shading, while absorbing large amounts of incident solar radiation and maintaining lower air temperature, which is further reduced by evapotranspiration from the trees. Open pools, fountains, ponds and running water were quite popular in the historical development of architecture, especially in southern dry climates. The phase change during water evaporation can decrease the dry bulb temperature of the air, though at the same time, there is an increase of the water content of the air.

Light coloured outer surfaces have been used extensively in traditional Greek architecture. The picturesque white villages in the Greek islands provide more than an aesthetically pleasing feeling. White surfaces reflect much of the incident solar radiation, thus reducing the heat transferred into interior spaces.

One of the most effective ways, however, of dealing with the problem of high temperatures during the day has also been the behavioural response of people. Use of different, cooler indoor spaces during the day or changes in building use altogether during summer, were very common behavioural actions. Gatherings in cooler open spaces, during the day or at the midday break, and outdoor sleeping during the night were also some simple ways of dealing with high temperatures and uncomfortable conditions.

Progress in science and technology has introduced tremendous changes in all fields, including the management of indoor conditions. In particular, advances in the fields of thermal sciences (thermodynamics, heat transfer, fluids) had as a result the design, development and production of mechanical systems capable of satisfying practically all needs in the field of cooling and refrigeration. The development of a practical cooling and refrigerating machine dates back to the middle of the nineteenth century, although there is evidence of the use of evaporative effects and ice for cooling in very early times [1]. Mechanical cooling for comfort got its start early in the twentieth century, but advances in this area have been rapid.

Mechanical cooling is achieved by several means [2], including:

- Vapour compression systems;
- Gas compression systems involving expansion of the compressed gas to produce work;
- Gas compression systems involving throttling or unrestrained expansion of the compressed gas; and
- thermoelectric systems.

Vapour compression is the most commonly used method in air-conditioning systems. The first equipment dates back to a British patent application by Jacob Perkins in 1835 [3]. The early systems were driven by steam engines and used ether as the working fluid. Compressors and overall equipment size became smaller as electric motors were substituted for the original steam engines and as vapour compression systems were applied to space air conditioning and commercial refrigeration.

Compression is accomplished mechanically or through absorption methods. The simple (theoretical) single-stage vapour compression refrigeration cycle is shown in Figure 1.1.

Cooling is accomplished by evaporation of the working fluid (a liquid refrigerant) under reduced pressure and temperature, in the evaporator, as a result of heat transfer from a high-temperature space. The refrigerant then enters the compressor as a slightly superheated vapour at a low pressure. It leaves the compressor and enters the condenser as vapour at some elevated pressure, where it is condensed as a result

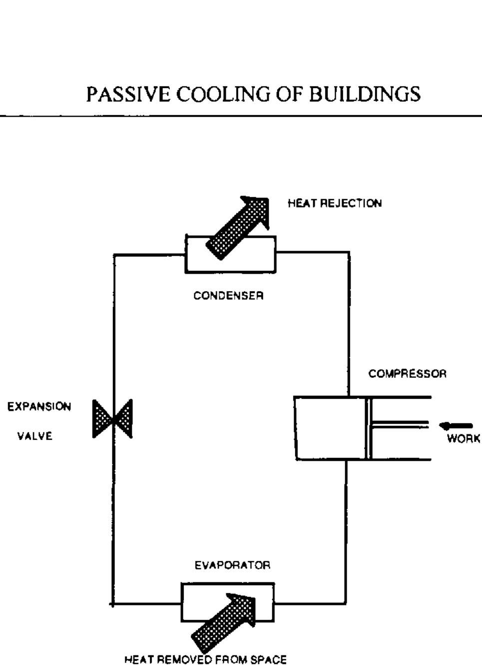

Figure 1.1 A simple vapour compression cycle

of heat rejection to ordinary cooling water or atmospheric air. The relatively high-pressure liquid is then throttled as it flows through the expansion valve. The thermodynamic cycle is completed as the remaining low-pressure liquid again enters the evaporator.

The most common refrigerants used in the 1920s and 1930s, were ammonia, carbon dioxide, methyl, ethyl, and methylene chloride, and isobutane, but they exhibited major disadvantages. Finally, the development of nontoxic, nonflammable working fluids, with acceptable operating temperatures and pressures and high efficiencies solved the problem for many decades. Eventually, CFC-12 was accepted as a standard working fluid for air-conditioning applications and refrigerators, HCFC-22 for residential air conditioning, CFC-11 for most commercial air-conditioning applications, and HCFC-502 for low temperature refrigeration.

Gas compression systems involving expansion of the compressed gas to produce work have found commercial applications in air refrigeration systems used for cooling aircraft spaces. The two gas compression systems are also used in the liquefaction of various gases.

Finally, there are some systems that produce cooling by thermoelectric means [4]. The thermoelectric device, such as a conventional thermocouple, utilizes two dissimilar materials. One of the junctions is located in the space under cooling and the other in ambient air, as shown in Figure 1.2. When a potential difference is applied, the temperature of the junction located in the cooled space decreases and the temperature of the other junction increases. Under steady-state operating conditions, heat is transferred from the cooled space to the cold junction. The other junction reaches a higher temperature than the ambient and, as a result, heat will be transferred from the junction to the surroundings.

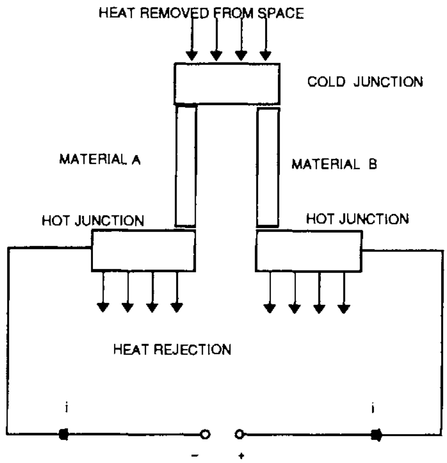

Figure 1.2 A thermoelectric device

Since in most cases there is a need both to cool and to heat a space, depending on the season, it appears that a system which can be used for both cooling and heating would be most attractive. This can be achieved with a heat engine, which is also known as a heat pump. The principle of the heat pump was introduced in the late mid-nineteenth century by Lord Kelvin. The system, however, received widespread application in the USA after World War II [5].

Technology advances that have provided systems with a high efficiency and performance and lower initial cost are described later in this chapter. However, the apparent concern for an overall reduction of energy consumption, including the increased cost of energy, and inherent problems of mechanical compression systems that operate with refrigerants which have been linked to atmospheric pollution, have caused a re-examination of alternative technologies and systems. Many of the previously described techniques of passive cooling, which have been applied successfully in the past, are now being revitalized, enhanced with new research findings, as well as the application of current technology and advanced system design know-how.

Published research has shown that a return to alternative energy sources, techniques and systems can be used to satisfy a major portion of the cooling needs in buildings. The trend, which started following the energy crisis in the mid-1970s, has stimulated a very active area of research with significant findings and great success in commercial applications.

This area has been named natural and passive cooling. The terms cover all naturally occurring processes and techniques of heat dissipation and modulation, as well as overheat protection and related building design techniques. This means without any other form of energy input than renewable energy sources or the use of other major mechanical systems. Included are several well known methods, enhanced with new capabilities allowed by technology advances, better understanding of the physical processes involved and optimum utilization of their potential effectiveness, in both traditional and modern applications, through a more comprehensive coupling and co-ordination of the available techniques and systems with the architectural design of the building in its environment.

Passive cooling techniques are also closely linked to thermal comfort of occupants. Indeed, some of the techniques used for passive cooling do not remove the cooling load of the building itself, but rather extend the tolerance limits of humans for thermal comfort in a given space.

It is also possible to increase the effectiveness of passive cooling with mechanically assisted heat transfer techniques that enhance the natural cooling processes [6]. Such applications are called 'hybrid' cooling systems. Their energy consumption is maintained at very low levels, but the efficiency of the systems and their applicability is greatly improved.

Passive cooling refers to techniques which can be used to [6, 7]:

- prevent heat gains and
- modulate heat gains.

Protection from or prevention of heat gains involves the following design techniques:

- microclimate and site design
- solar control
- building form and layout
- thermal insulation
- behavioural and occupancy patterns and
- internal gain control

while heat gain modulation can be achieved by proper use of the building's thermal mass (thermal inertia), in order to absorb and store heat during daytime hours and return it to the space at a later time.

Natural cooling refers to the use of natural heat sinks for excess heat dissipation from interior spaces, including :

- ventilation
- ground cooling
- evaporative cooling and
- radiative cooling.

By combining different passive and natural cooling techniques, it is possible to prevent overheating problems, decrease cooling loads and improve comfort conditions in buildings.

The principles governing each of the above topics are briefly presented in the following discussion, for each major category. A more comprehensive treatment is presented in other chapters of this book. The majority of these methods were extensively investigated in the USA and Israel following the major oil crisis in the mid-1970s. In Europe, similar research was initiated in the mid-1980s. A comprehensive overview of the current state of passive cooling research is available in [8–10].

Passive cooling by heat gain prevention

The microclimate and proper site design can greatly influence the thermal behaviour of a building. The overall principle is that a building must be adapted to the climate of the region and the microclimate (that is, the immediate environment around a building). The site design is influenced by economic considerations, zoning regulations and adjacent developments, all of which can interfere with the design of a building with regard to the incident solar radiation and the available wind. Vegetation does not only result in pleasant outdoor spaces, but can also improve the microclimate around a building and reduce the cooling load. It has been estimated that a full size tree evaporates 1,460 kg of water on a sunny summer day, which is equivalent to 870 MJ of cooling capacity [11]. Areas with high vegetation may exhibit noticeably lower air temperatures (by 2 to 3°C).

Solar control is the primary design measure for heat gain protection. The use of various shading devices in attenuating the incident solar radiation as it enters into the building [12], can significantly reduce the cooling load. External shading of opaque walls with surrounding natural vegetation, external or internal shading and high performance glazing can result in satisfactory thermal and optical performances. This subject is treated in Chapter 10.

The building form and internal space layout determines the exposure of interior spaces to incident solar radiation, as well as to daylight and wind [6]. The building shape controls both heat losses and heat gains, by reducing or increasing the ratio of the exposed surface to the volume. It is determined by planning regulations, space availability, neighbouring site development, architectural styles, client preferences and cost constraints.

The building envelope determines the physical processes taking place between the outdoor environment and indoor spaces. The objective is to limit thermal gains during the summer, due to high outdoor air temperatures and incident solar radiation. Thermal insulation can reduce the heat conducted through the building materials. During summer it reduces thermal gains and during winter it reduces thermal losses. The level of thermal insulation in buildings is determined by national codes and is mandatory in most countries. An increase of insulation above recommended values results in a very small decrease in the cooling load.

Behavioural and occupancy patterns in a building can be properly adjusted in order to achieve thermal comfort and consequently reduce the energy consumption for cooling. Dressing according to the prevailing weather conditions, adjusting the levels of physical activity, moving to cooler spaces in the building, adjusting thermal controls (opening and shutting windows, use of blinds and curtains for shading) are simple but yet effective actions [13].

The control of internal gains can be achieved by proper design and operation of internal heat sources, such as artificial lighting, equipment and occupants. For example, use of energy-efficient lamps can reduce heat dissipation into the space, in addition to reducing electrical energy consumption. Human activity can greatly influence internal heat gains, but it is difficult to modify, other than by appropriate zoning within the building's spaces. Accordingly, groups of people with similar activities should be placed in the same area in order to be able to best satisfy their needs. Equipment which requires special air conditions, such as computers or other electronic equipment, and spaces with high equipment concentration must be taken into account when calculating the cooling load. Similarly, a large number of occupants in a given space will increase the latent heat of the space and the amount of air ventilation required. In general, internal heat loads must be carefully accounted for, especially in large buildings that extensively utilize artificial lighting, in spaces that are usually heavily equipped and in spaces that are occupied by a large number of people. In fact, large office buildings require some cooling almost all year round. The methodology for cooling load calculations is presented in Chapters 7 and 14.

Passive cooling – heat gain modulation

The thermal mass of a building (typically contained in walls, floors and partitions, constructed of material with high heat capacity) absorbs heat during the day and regulates the magnitude of indoor temperature swings, reduces peak cooling load [14] and transfers a part of the absorbed heat into the night hours [15]. The cooling load can then be covered by passive cooling techniques, since the outdoor conditions are more favourable. An unoccupied building can also be pre-cooled by ventilation during the night and this stored coolness can be transferred into the early hours of the following day, thus reducing energy consumption for cooling by close to 20% [16]. The role of thermal mass in reducing and regulating the cooling load and indoor air temperatures is treated in more detail in Chapter 8.

Natural cooling – heat sinks

Natural or forced ventilation is one of the primary means of reducing the cooling load in buildings (removing heat from indoor spaces) and of extending indoor thermal comfort conditions for humans when the outdoor conditions (temperature and humidity) are favourable. Ventilation is also necessary in all indoor spaces, in order to introduce the required levels of fresh air and to control odours and indoor pollutants. Quantified ventilation research has been conducted continuously since the 1930s [17].

Natural ventilation is caused by pressure differences at the inlets and outlets of a building envelope, as a result of wind velocity and/or stack effects. To enhance its effectiveness, the wind can be properly channelled and moved through the building. Night ventilation, wind towers and solar chimneys are the main natural ventilation techniques. The subject of passive cooling with natural ventilation is also addressed in the penultimate section of this chapter and treated in more detail in Chapter 9.

Forced ventilation is achieved by mechanical means, using fans to achieve and control the appropriate airspeed. Ceiling, attic or simple portable room fans can be used to control the indoor air movement and achieve the appropriate air changes. Ceiling fans allow for higher indoor air temperatures, since increased indoor air movement enhances occupants' thermal comfort conditions (Chapter 6). As a result, cooling load requirements are reduced, with reductions ranging between 28% and 40% of the cooling costs [18], depending on the climate. Ceiling fans have been proved to be a viable technology and their low capital, operational and maintenance costs make them especially attractive.

Ground temperatures remain almost constant during the day at depths exceeding 1 m, while there are small seasonal changes which extend to a depth of 9 to 12 m. Even during hot summer days, ground temperatures remain significantly lower than ambient air temperatures, with small daily variations compared to the diurnal cycle of ambient air temperature and solar radiation [19]; this is due to the high thermal capacity of the soil. Accordingly, it is possible to couple the building with the ground for cooling purposes [20], either indirectly or directly. Indirect ground coupling involves a hybrid system. Indoor air is circulated through underground pipes, to dissipate heat to the lower-temperature heat sink and then the cooled air is returned to the building. A similar concept can be traced to the mid-16th century in Italy [21] and to ancient Greece on the Aegean island of Delos. The performance of buried pipes depends on the inlet air temperature, ground temperature, air circulation rate and pipe characteristics such as the thermal properties of the materials, dimensions and depth [22].

Another way to take advantage of the lower ground temperatures is by placing the building in direct contact with the ground [23]. In early human history, underground shelters were used extensively, utilizing the ground as the first insulating material. Today, partial or total underground construction is a viable alternative to conventional architectural styles, because of the resulting energy conservation associated with such a construction, especially in climates with extreme ambient conditions [19]. In Europe, there are a large number of one- or two-storey buildings partially buried in hillsides, with reported energy savings of 50 to 90% [10]. Earth-sheltered buildings can also provide isolation from noise and air pollution, although there are several design and operational difficulties that need to be taken into account.

Evaporative cooling applies to all processes in which the sensible heat in an air stream is exchanged for the latent heat of water droplets or wetted surfaces [24]. Warm outdoor air comes into contact with water droplets, which can be sprayed directly into the air stream, or it is passed through a wetted porous material. The moisture evaporates, thus extracting heat from the warm air and lowering its temperature. The majority of evaporative cooling applications use hybrid systems. Small mechanical systems, like pumps and fans, are necessary for moving the fluids (air and water) used during the process. When the water evaporation occurs in direct contact with the ambient air this is a direct process. The problem with such a process is that it increases the moisture content and consequently it cannot be used in regions with high humidity levels. In such a case, it is possible to use an indirect process, in which air is cooled without addition of moisture by passing through a heat exchanger, with a consequent reduction in efficiency. Evaporative cooling techniques were best demonstrated at the Seville Universal Exhibition (EXPO'92) in Spain [25–29].

Cool towers can also be used to move air masses through interior spaces and eliminate the use of blowers and its electrical cost. In a prototype application, a cool tower was fitted to a test house at the Environmental Research Laboratory, University of Arizona, Tuscon [8]. The air circulation achieved by the mechanism inside the house was 2.7 air changes per hour. Maximum air temperatures inside the house did not exceed 26°C, during the summer period. Traditionally, cool towers have been used in Middle Eastern architecture [30] and in natural-draught evaporative cool towers [31, 32], by moving ambient air through evaporative cooling pads, resulting in a lower air temperature at the exit of the tower. In a similar approach [33], the towers use fogging devices (atomizers) as the cooling source of the system, which allows for better regulation relative to the prevailing conditions and minimization of air-pressure losses.

Radiative cooling is a technique which can be used either passively or as a hybrid technique [34]. It is based on the fact that every object, being at a temperature higher than 0°K, emits energy in the form of electromagnetic radiation. If two elements at different temperatures are facing each other, a net radiant flux will occur. In the event that a low-temperature element is kept at a constant temperature, then the other element will radiate and cool down, in order to reach an equilibrium with the colder object. In the case of building radiative cooling, the building envelope (or another appropriate device such as a metallic flat-plate radiative air cooler) is cooled by dissipating infrared radiation to the sky, which acts as a low-temperature environmental heat sink. The amount of radiant exchange depends on the temperature difference between the sky temperature and the building element. Clear skies exhibit low sky temperatures, while cloud cover, air humidity and pollution decrease the effectiveness of radiative cooling processes [35].

The roof is the most important passive radiative cooling system in a building, because it continuously faces the sky dome. Roof colour influences the thermal performance of a building because it governs the absorption and reflection of incident solar radiation during daytime and the emission of long-wave radiation during night time, especially for light structures [36]. Measurements of the potential of this technique [37] give a cooling potential of 0.014 kWh m^{-2} per day. To enhance the performance of this technique, it is possible to protect the roof during the day with a movable insulation system and expose it during the night. Operable insulation can be in the form of horizontal movable panels or hinged panels, positioned vertically during the night. Measurements of the performance of this technique give a cooling potential of 0.266 kWh m^{-2} per day [37]. Alternatively, it is possible to use water bags placed on the roof, a system known as the Skytherm [38]. This is the only system that, with favourable climatic conditions, can provide both cooling and heating.

The potential of radiative cooling for 28 locations around southern European countries has been simulated and evaluated in [39] for a flat-plate radiator (hybrid system). A comparison has also been performed with the results from four localities in south-eastern USA. Accordingly, it has been concluded that southern Europe exhibits a promising potential for the use of this technique. The estimated mean daily useful cooling energy delivered by a flat plate radiative cooler ranges between 55 and 208 Wh m^{-2} for average sky conditions, and between 68 and 220 Wh m^{-2} for clear sky conditions, depending on the location. For areas like the southern USA, with high humidity levels, the potential effectiveness is limited, ranging between 41 and 136 Wh m^{-2} for average sky conditions and between 69 and 182 Wh m^{-2} for clear sky conditions. Some results on specific aspects of radiative cooling have also been reported in [40–42].

Comparative information regarding the thermal performance for a typical building of several passive and hybrid cooling techniques involving the use of natural cooling techniques, such as ground cooling, direct and indirect evaporative coolers and night ventilation techniques are available in [43]. The calculations were performed for a typical 80 m^2 one-storey dwelling located in Athens, Greece. Accordingly, the results have shown that :

- Night ventilation techniques can provide a part of the building's cooling load, being more effective during the months with lower night outdoor air temperature (June and August). The maximum depression of the peak indoor temperature does not exceed 1°C;
- For ground cooling, the earth-to-air heat exchangers have to be buried at a depth ranging between 3.5 and 5 m, with peak indoor air temperature reductions ranging between 2 and 5°C, respectively. The diameter and air velocity variations were found to affect the indoor air temperature significantly. For example, an increase of the diameter from 0.20 m to 0.22 m results in an indoor temperature decrease of 1.5°C.
- The use of a direct evaporative cooler, a parallel-plate-pad evaporative cooler with a 50 m^2 wetted area, can reduce the maximum peak indoor air

temperature by 4 to 6°C. The indoor temperature decrease is satisfactory for a fan rate higher than 1,500 r.p.m.

- The use of indirect evaporative coolers can lead to acceptable indoor temperature levels for an air speed through the cooler of 0.1 m s^{-1}. During the hot month of July, an air speed of 0.3 m s^{-1} is recommended.

BUILDINGS AND ENERGY CONSUMPTION FOR COOLING

The total primary energy consumption in the European Community in 1990 was 725 million tons of oil equivalent (mtoe) [44]. Transportation and industry each consumed 220 mtoe, while for domestic and tertiary uses the total is 275 mtoe. The final energy consumption in the domestic/tertiary sector, for each member state, is given in Table 1.1. Oil with a 44.3% share, was the primary energy source, while solids represented 23.8%, natural gas 17.5%, nuclear 12.9% and all other technologies, including renewable energy sources, were only utilized to satisfy 1.6%.

Table 1.1 Final energy consumption in the domestic/tertiary sector in the EEC member states for 1990

Member state	Energy consumption [mtoe]
Belgium	12.0
Denmark	5.8
France	51.0
Germany	68.0
Greece	4.0
Ireland	3.0
Italy	38.2
Luxembourg	0.7
Netherlands	19.0
Portugal	2.3
Spain	13.5
UK	57.5

During the last decade, Belgium, Denmark, Germany and The Netherlands have achieved a small energy consumption decrease, France, Luxembourg and the UK have maintained a constant average, while there has been an increase in Greece, Ireland, Italy, Portugal and Spain. Clearly, further actions are necessary. In 1989 the European Commission adopted a programme of urgent measures to reinforce and expand efforts in energy-efficient improvements, in energy conservation, combined with the use of more non-fossil-fuel energy sources, and in the protection of the environment.

Overall, energy consumption in commercial and residential buildings represents approximately 40% of Europe's energy budget [45]. This is allocated for satisfying the buildings' energy needs for heating, cooling, lighting and the other electrical appliances and equipment.

Passive solar design reduces fossil and nuclear fuel consumption for cooling in buildings. According to a study on the current and future use of passive solar energy in buildings in the European Community [46], there is a large potential for considerable energy savings in cooling of buildings, especially in southern regions. In Italy, for example, during 1990 the solar energy contribution to non-domestic cooling was 2.3 mtoe and 5.2 mtoe for domestic cooling purposes. In 2010, there is a technical potential to reach 5 mtoe and 12 mtoe, respectively. The consequent reduction in atmospheric pollution in 2010 is estimated to reach 3 million tones of CO_2 for domestic cooling and 0.01 million tones of SO_2. Similar information for other south European countries is shown in Figure 1.3.

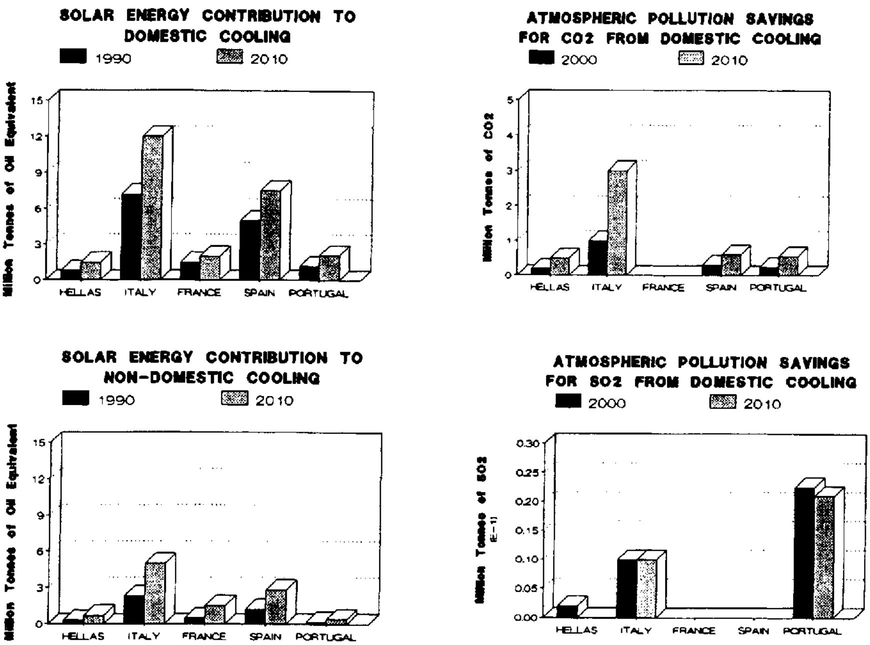

Figure 1.3 Contribution of solar energy to cooling and consequent reduction in atmospheric pollution in southern European countries [46]

Energy consumption for cooling of building in southern countries is of primary concern. A recent study carried out in Greece [47], a country with a representative south European climate, has provided a comprehensive picture of the energy consumption in buildings. These energy audits were performed in over 1,000 public and commercial buildings, including air-conditioned and naturally ventilated office, commercial, school, hospital and hotel buildings, around the country. Each type of building exhibits individual characteristics and problems due to specific operational needs, and therefore requires independent treatment and analysis.

Accordingly, the average annual final energy consumption for all the audited buildings in the five categories is 187 kWh m^{-2} in offices [45], 152 kWh m^{-2} in commercial buildings [48], 93 kWh m^{-2} in school buildings [49], 406.8 kWh m^{-2} in hospitals [50] and 273 kWh/m^2 in hotels [48].

To appreciate the impact of mechanical air-conditioning systems on energy consumption, one should refer to Table 1.2. The total consumption and the annual average final energy consumption for cooling in air-conditioned (A/C) and naturally ventilated (N/V) buildings is given for each one of the five building categories. In offices cooling represents 12.6% of the total average annual final energy consumption, in commercial buildings 12.2%, in schools only 2%, in hospitals less than 1% and in hotels about 4%. As shown in Table 1.2, air-conditioned buildings consume considerably more energy than naturally ventilated buildings, ranging between 4% to over 54%, depending on the type of building.

Table 1.2 Annual average values of total final energy consumption in buildings in Greece

	A/C Buildings		**N/V Buildings**	
Building Category	**Number of Buildings**	**Total Energy [kWh/m²]**	**Number of Buildings**	**Total Energy [kWh/m²]**
Offices	153	226	33	179
Commercial	261	215	254	206
Schools	16	183	150	119
Hospitals	26	352	6	236
Hotels	78	285	62	270

In comparison, typical primary energy values for offices in northern climates range between 270 to 350 kWh m^{-2} [51]. The electrical energy consumption in mechanically air-conditioned buildings ranges between 93 to 115 kWh m^{-2} for school buildings [52], while for a newly constructed 25,000 m^2 office building (with laboratory facilities and heavily equipped spaces) the electric energy use amounted to 214 kWh m^{-2} [53].

Information from a study in France has provided similar results [54]. For offices, the annual average total energy consumption ranges between 145 kWh m^{-2} in small communities (population less than 2,000 inhabitants) and 165 kWh m^{-2} in large cities (population greater than 50,000 inhabitants). For schools, the annual average total energy consumption ranges between 147 kWh m^{-2} in large cities and 175 kWh m^{-2} in small communities. For social and cultural buildings the corresponding energy consumption ranges between 77 kWh m^{-2} in small communities and 188 kWh m^{-2} in large cities.

The cumulative distribution of the total final energy consumption in each category of the audited buildings in Greece, is shown in Figure 1.4.

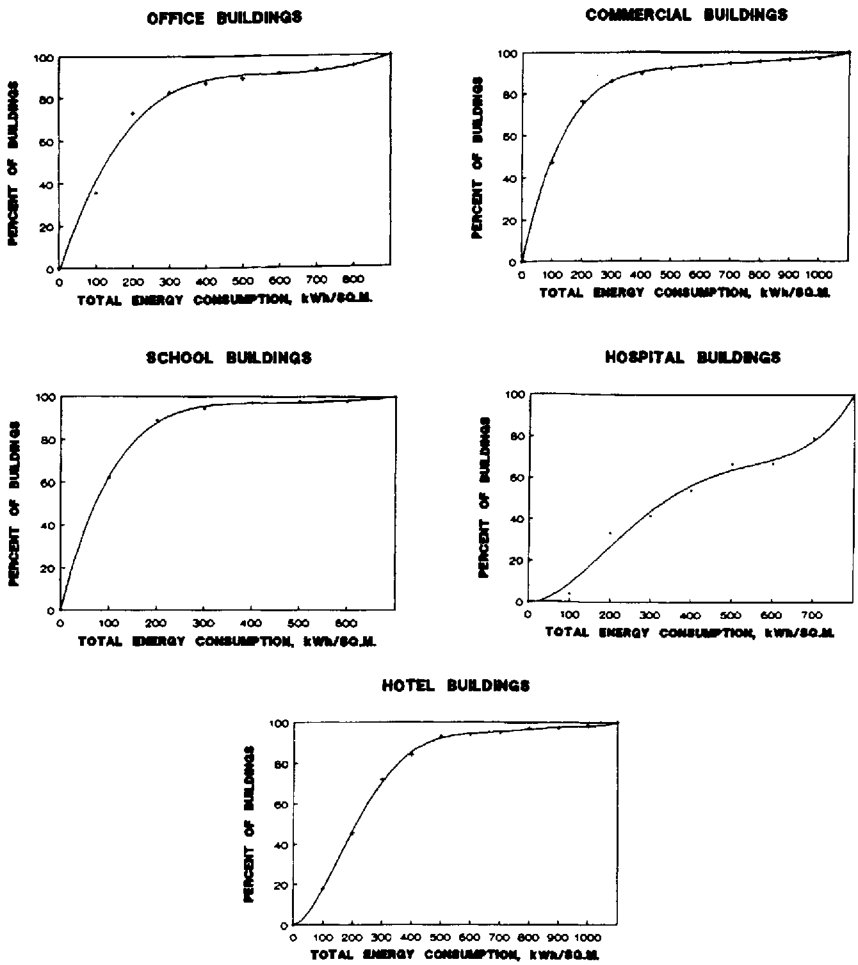

Figure 1.4 Cumulative distribution of the total energy consumption in buildings

Office buildings exhibit an annual average total energy consumption of 187 kWh m^{-2}. The minimum value from the collected data was 6 kWh m^{-2} and the maximum value 2,100 kWh m^{-2}. Approximately 39% of the office buildings have an energy consumption less than 100 kWh m^{-2}, while for 36% of the buildings their energy consumption ranges between 100 and 200 kWh m^{-2}.

Commercial buildings have an annual average energy consumption of 204 kWh m^{-2} (with for the audited buildings a minimum value of 3 kWh m^{-2} and a maximum value of 6,400 kWh m^{-2}). About 48% of the buildings consume less than 100 kWh m^{-2}, while for 32% the values range from 100 to 200 kWh m^{-2}.

Schools operate for nine months and remain closed during the summer months. Some schools also operate only during the morning hours and, as a result, the total energy consumption has an annual average value of 110 kWh m^{-2} (a minimum value of 6 and a maximum value of 900 kWh m^{-2}). Accordingly, 27% of the buildings consume less than 100 kWh m^{-2}, while 47% have an energy consumption ranging between 100 and 200 kWh m^{-2}.

Hospitals exhibit the highest annual average total energy consumption (407 kWh m^{-2}), as a result of their continuous 24-hour operation and the use of a large amount of health-care equipment. The actual data ranged between 100 and 1,400 kWh m^{-2}. About 33% of the buildings consume less than 200 kWh m^{-2}, while 21% of the buildings are in the range of 200 to 400 kWh m^{-2}.

Hotels exhibit the second highest annual average total energy consumption, equal to a value of 273 kWh m^{-2}. Among the audited buildings, the minimum value was 20 kWh m^{-2} and the maximum value was 1,500 kWh m^{-2}. Approximately 19% of the hotels have an energy consumption less than 100 kWh m^{-2}, while for 33% of the buildings the consumption ranges between 100 and 200 kWh m^{-2}.

The impact of air conditioning on the total energy consumption of a building can be significant. For example, in N/V and A/C office buildings the annual average total energy consumption is 179 kWh m^{-2} and 226 kWh m^{-2}, respectively. Consequently, the use of A/C in office buildings increases the annual energy consumption by an average value of 40 to 50 kWh m^{-2} [45]. The distribution of electrical consumption of the N/V and A/C buildings, in each one of the five categories, is shown in Figure 1.5.

For offices, 42% of the N/V and 38% of the A/C buildings exhibit an annual average electrical energy consumption lower than 100 kWh m^{-2}. In addition, 87% of the N/V and 84.7% of the A/C buildings present an electrical consumption lower than 300 kWh m^{-2}.

For commercial buildings, the annual average electrical energy consumption is less than 100 kWh m^{-2} for 58% of the N/V and 38% of the A/C buildings, while for 25% of the N/V and 35% of the A/C buildings the consumption is between 100 and 200 kWh m^{-2}.

For schools, 27% of the N/V buildings and 29% of the A/C buildings consume, on an average annual basis, less than 100 kWh m^{-2} of electrical energy, while 49% of the N/V and 35% of the A/C buildings have a consumption between 100 and 200 kWh m^{-2}.

For hospitals, over 50% of the N/V and only 5% of the A/C buildings have a consumption less than 100 kWh m^{-2}, while 25% of the N/V and 29% of the A/C exhibit a consumption ranging between 100 and 200 kWh m^{-2}.

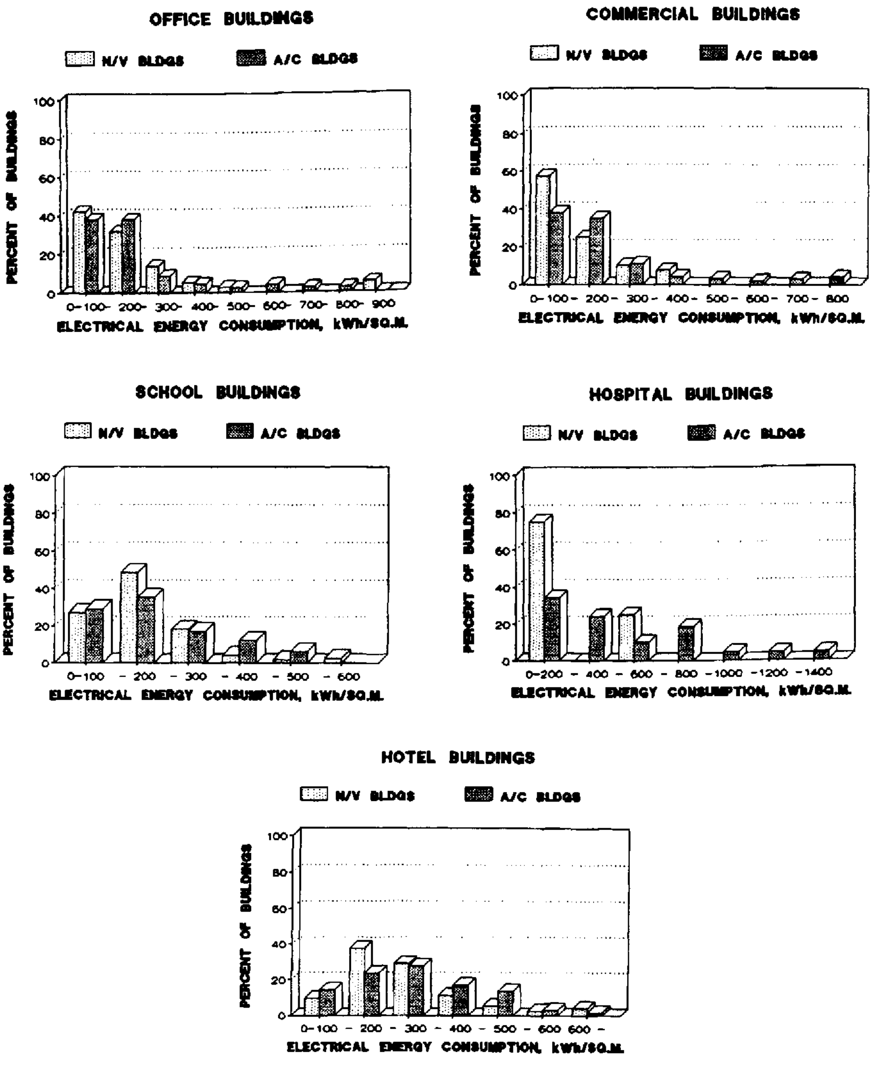

Figure 1.5 Distribution of annual average electrical energy consumption of naturally ventilated (N/V) and air-conditioned (A/C) buildings

For hotels, 47.2% of the N/V and 38% of the A/C buildings have an annual average electrical energy consumption less than 200 kWh m^{-2}, while 40.3% of the N/V and 45% of the A/C buildings have a consumption between 200 and 400 kWh m^{-2}.

Overall, this information, along with other related guidelines [55], can prove very useful for efficient energy planning and building design. Following a series of simulations and analysis, it has been shown in [47] that it is possible to reach an overall 20% energy conservation. In particular, several interventions in the buildings, to reduce the energy consumption for cooling, were investigated and assessed. Accordingly, it is possible to achieve the following energy savings for cooling by:

- reducing external loads (proper shading can result to a 30% reduction);
- reducing internal loads from lighting by using fluorescent lamps (80 lm/W) can reduce the cooling load by an average of 10%;
- using natural cooling techniques (like ground, evaporative cooling), where possible;
- using night ventilation techniques, with six air changes per hour, can result in substantial energy savings for cooling, up to 80%, depending on the type of building; and
- using ceiling fans can result in a very high percentage of energy savings for cooling, up to 95%, depending on the type and function of the building.

Significant energy savings for cooling, but also for heating, purposes can be achieved by using control strategies of the indoor air temperature [56]. Indoor air temperature is controlled in such a way as to drift from the optimum set-point to conditions which are still acceptable for most occupants. In office buildings this can result to energy savings up to 10% [57]. In another recently completed study of a large office building [58], indoor air temperature was allowed to increase in the afternoon hours (after 3 pm). Simulation results have shown that it is possible to achieve a reduction of seasonal consumption of chilled water of between 34 and 40% and a reduction in the energy budget for heating, ventilating and air-conditioning systems of 11%.

Finally, it is important to consider an additional parameter for energy savings in existing mechanical cooling systems. Proper maintenance of air-conditioners can reduce electrical peak demand, save energy and consequently reduce operating costs, according to a pilot project recently completed in the USA [59]. Savings and peak reduction estimates for individual repair measures such as correcting low airflow rates (due to clogged-up filters), repairing overcharging and undercharging of the refrigerant and repairing duct leakage can achieve 8 to 18% savings in cooling.

CURRENT TECHNOLOGY OF AIR-CONDITIONING SYSTEMS

Air-conditioning systems are used to control the temperature, moisture content, circulation and purity of the air within a space, in order to achieve the desired effects on the occupants of the space or on the products and equipment manufactured or stored there. Year-round A/C includes summer cooling (maintaining the indoor temperature below that of the outdoor air), winter heating (maintaining the indoor temperature above that of the outdoor air), dehumidification (lowering the water content of the ambient air to acceptable levels, usually during the cooling season), and humidification (addition of water vapour to the indoor air, usually during the heating season).

Air-conditioning systems are grouped into two main categories, namely central units and independent units.

Central air-conditioning systems

High-capacity central A/C systems are primarily used in large buildings. The main unit is usually placed in a mechanical room, generally at a distance from the conditioned spaces. The central unit is connected, through a duct/piping system, to individual units which are installed in each air-conditioned space. The outdoor air is introduced into the main A/C unit and is mixed with an amount of recirculated air. The mixture then passes through air filters, in order to remove any dust or other foreign particles.

The air is then conditioned, according to the operating mode (cooling or heating) of the system. For cooling purposes, the air is cooled and if necessary is also dehumidified. For heating purposes, the air is pre-heated by passing through a system supplied with steam or hot water, if necessary water vapour is also added passing through a humidifier, and finally the air is heated again by using steam or hot water. The air is then transported with fans (at speeds ranging between 5 and 15 m s^{-1}) through a duct system, usually at ceiling height, and is then diffused and circulated inside the conditioned spaces. There are various types and designs of inlet–outlet air diffusers, which are usually placed at ceiling height, floor level or near the windows, in order to ensure the best possible air circulation in the space.

For applications where there are zones in the building which may require warm and cool air at the same time, it is possible to use two independent air-duct systems, one for each air stream. In this way the air supplied into each zone of the building can be independently controlled to meet specific requirements. A far more costly design of several air duct systems is also necessary in some buildings, in order to supply air in significantly different conditions. For example, computer rooms require a year-round constant temperature of 20°C, while working environments may require temperatures during summer of 25–26°C.

Finally, depending on the type of building, an appropriate percentage of indoor air is transferred back to the central unit, for mixing with fresh but not conditioned outdoor air. This results in considerable energy savings compared to using and conditioning 100% outdoor air. Sometimes this is mandated by the type of building, as in the case of hospitals. Central A/C systems can be sized accordingly, in order to satisfy the cooling and heating loads of the building.

Rather than use all-air systems, it is also possible to use air–water systems. One possible way is to partially condition the air at the central unit, transfer it to the individual spaces, where the air is treated in a heat exchanger by using either a hot or cold water stream, in order to achieve the desired temperature. These systems require, in addition to the air ducts, a pipe system for the circulation of water and also a system for the heating or refrigerating of the water. These systems allow the occupant to control and easily achieve the desired conditions inside the space.

To reduce the cost of the air–water units, it is possible to exclude the air circulation system from the central unit. Instead, the local system in each conditioned space is supplied with hot or cold water, while the indoor air is circulated by a fan in the system, and passed through a heat exchanger where it is heated or cooled. These systems are known as fan-coil units and can be placed on the floor or at ceiling height.

Some innovative elements which have proved successful in the operation of central systems include the use of cogeneration concepts for refrigeration prime movers and waste energy recovery using absorption refrigeration for subcooling to boost the coefficient of performance of the refrigeration plant [60].

For large scale applications, chilled water storage may also be used in order to increase the performance, reliability and energy efficiency and to lower the operational cost of a central cooling system. A partially buried 10,000 m^3 chilled water storage reservoir was coupled with a 4200 ton central chiller plant, in order to reduce the annual cooling electricity usage for a 102,000 m^2 defence electronics manufacturing facility [61]. The thermal energy storage system reduced net annual cooling electricity usage at the facility by 970 MWh (representing 9% of the total energy consumption) during the first year of operation and by 3 GWh (about 28.3%) during its second year of operation, in comparison to a conventional cooling system.

Independent air-conditioning systems

Smaller, independent A/C systems can be placed in any space without the need of a central system. They are mostly used in buildings where occupants desire to install air conditioning easily and quickly in selected spaces only, or in older buildings where the renovation cost for installing central systems is not justified. Usually, independent A/C systems are small units, since they are only used for satisfying the loads of small spaces, typically ranging between 9000 and 25,000 Btu/hr.

There are primarily two types of systems, namely monoblock and split units.

Monoblock units are independent air-conditioning systems that use older design and technology. All the components are placed in the same housing. They are usually placed inside the structural material of an exterior wall of the space (in a hole of the size of the unit) or in some cases they are placed at a window that has been properly modified.

Split units offer easier installation, since they do not require any major structural changes of the space's construction material. One part of the unit is placed outside (housing the heavy mechanical parts – compressor and a fan), while another part, containing the evaporating unit and a blower, is placed anywhere in the indoor space. The two units are connected by two small-diameter pipes for transferring the working fluid. Indoor air is circulated through the indoor unit, passes through the heat exchanger and then through a filter, before it is returned into the space. Operating noise levels are very low (since the main mechanical parts are placed outdoors), while the indoor units are designed to be aesthetically pleasing, in various sizes and with various controls (thermostats, operating timers, fan speeds, telecontrols). The indoor units can be easily mounted on the wall or ceiling, or placed on the floor, depending on the needs of the occupant.

Heat pumps

The most popular current technology for small to medium cooling/heating loads is the heat pump. Air conditioners and heat pumps were first used extensively in Japan, but they have now penetrated the European and US markets. The characteristics of these systems are analysed in the following discussion.

Any refrigeration system acts, to an extent, as a heat pump. Heat is transferred from a low-temperature space and is rejected at a higher temperature level. If the system is only to produce a cooling effect, then the rejected heat is usually wasted. In a heat pump, the rejected heat in the condenser is utilized to heat a space during the periods that heat is required.

A simple schematic diagram representing the operation of a heat pump is shown in Figure 1.6.

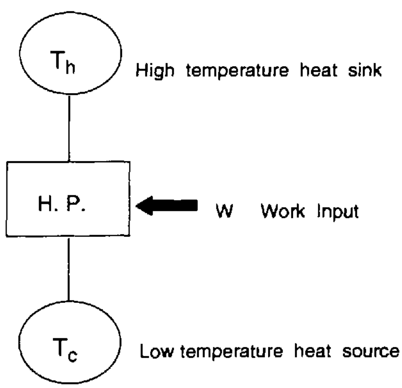

Figure 1.6 Basic heat pump operation

Heat pumps are categorized according to the type of operation of the compressor, as electrically driven or as an internal combustion. Depending on the location of the various components of the heat pump system, they are identified as monoblock units (when all components are placed in the same housing) or split units (when the condenser is separated from the remaining system). Depending on the type of the heat source for the evaporator (air, water, ground) and the heat sink (air, water), they are identified as air-to-air, air-to-water, water-to-water, water-to-air and ground-to-water heat pumps. Of these, outdoor air and well water are the most commonly used, working as a reversed Carnot cycle (smaller size and economical operation).

Heat pumps operate most efficiently with small temperature differences between the heat source and heat sink ($T_h - T_c$). Consequently, one must look for a heat source with the highest possible temperature, which has to be lower than the desired one; otherwise there is no need for the heat pump. Heat pumps are available in various sizes, small (6000 to 25,000 Btu/hr), medium size (25,000 to 100,000 Btu/hr) and large units (100,000 to 600,000 Btu/hr).

Small, split-system heat pumps with variable speed drives are becoming quite popular throughout the world, owing to their ease of installation, quiet operation and energy savings. During 1990, over 30% of all home air conditioners sold had a variable speed feature [62]. The electrical energy savings of variable-speed systems can be significant, resulting in reductions ranging between 10 and 30%. The variable speed permits the air-conditioner to maintain high efficiency while operating at part load. To achieve this, systems utilize an electronic chip called an inverter.

The inverter converts the mains frequency (i.e. 50 or 60 cycles per second) alternating current into direct current and then back to alternating current with a frequency of 30–120 cycles per second, depending on the cooling load. The motors can then vary the speed over a wide range of heat removal rates. Because of the power consumption of the inverter, however, a power loss ranging between 7 and 10% has been reported. At maximum cooling load conditions, this can even reach 35%.

Mechanical compression systems using vapour refrigerants are predominant among the cooling methods. All the mechanical cooling systems operate in a cycle that requires work and accomplishes the objective of transferring heat from a low-temperature system to a high-temperature system. In order to assess the effectiveness of the various thermodynamic systems, it is possible to use a dimensionless parameter, called coefficient of performance (COP), which is defined as the ratio of the useful effect to the net energy supplied from the external source. For a heat pump that operates in the cooling mode, the useful cooling effect is (Q_L), the heat transferred from the cooled space, while the energy that costs is the work (W) and is supplied from external sources. Accordingly, for the theoretical single-stage cycle of Figure 1.6, the coefficient of performance is given by:

$$\mathrm{COP} = Q_L / W = Q_L / (Q_H - Q_L) = 1 / (Q_H / Q_L - 1)$$

The COP of the various systems is a determinant factor on the final energy consumption of the unit. An increase, for example, of the coefficient of performance (for example by using a heat pump with a COP of 2), can reduce the electrical energy consumed by the cooling system, by as much as 25%.

Problems and prospects in the A/C industry

During recent years, the use of mechanical air conditioners in southern European countries has increased significantly. This is primarily due to an increase in the living standards in these countries and a reduction in the cost of A/C units. Figure 1.7 shows that there is a clear trend of increasing sales with gross national product (GNP) in European Community member states [63]. In Greece, the sales of A/C units showed an unprecedented increase during the late 1980s due to a series of heat waves over a period of three years (Figure 1.8). Sales of A/C units in Greece, have increased by 900% during this period [46].

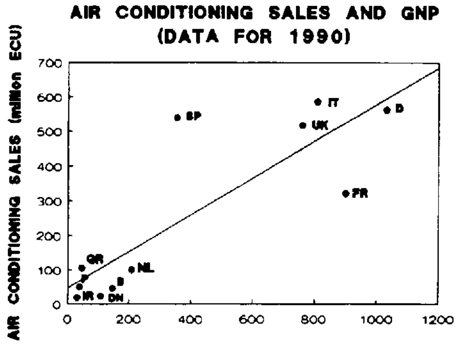

Figure 1.7 Sales of air conditioners as a function of Gross National Product in European Community member states [63]

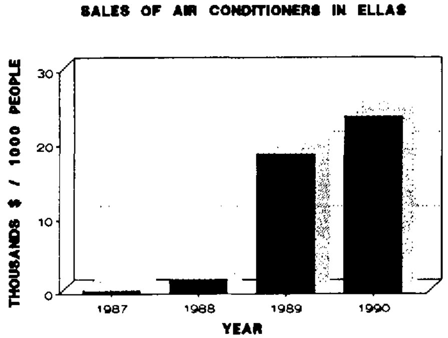

Figure 1.8 Sales of air-conditioners in Greece during the period of 1987 to 1990

Unfortunately, as a result of the sudden increase in demand for air-conditioning systems and primarily for small split units, the Greek market was invaded by a number of low-quality, low-efficiency, but also low-priced, units. The systems were primarily old-technology air-conditioners, not designed for the prevailing climatological conditions in Greece. However, customers looking for a low-price bargain purchased and installed thousands of these systems. The fact that there were no official standards or relavant legislation meant that there was no protection for the consumer.

The negative impact was immediate. Bad performance, high operating costs and numerous malfunctions of the systems started a domino effect. People lost confidence in air-conditioning systems and the market exhibited a decline. The positive result was that most of the market invaders were eventually forced out of business. The negative overall result, though, was that people who had made an investment, in their attempt to justify it, continued using the systems. In addition to all this, the properly operating units and the continuous promotion of sales by the well established major air-conditioning companies created a more serious, and, it appears permanent, negative effect.

Clearly, the impact of this increase on electrical energy consumption has been alarming. During this period, the peak electric energy load in Greece occurred, for the first time, during the summer [64]. Similar trends have also been observed in most southern regions in Europe, the USA [65], Japan and the Middle East. During the summer of 1990, Tokyo Electric Power Company, the world's largest private utility, experienced a summer peak only 1% below its maximum available generation capacity [62], due primarily to the increasing popularity of air-conditioner systems. In Kuwait, air-conditioning systems consume 75% of the total electrical energy consumption during the summer months [66].

This can potentially create severe problems in the availability of electric energy, especially in countries where the electrical energy production is primarily based on hydroelectric power plants, as in Greece. Hydroelectric power plants do not operate up to their potential during the dry summer months. Consequently, in order to avoid possible energy shortages, if the same trends in energy consumption continue, power companies must build new power generation plants to satisfy these short seasonal needs. This will clearly require significant financial investment, which is difficult to justify if a plant is to be used only during the summer period. Furthermore, if thermal plants are considered, this will have a direct impact on atmospheric pollution. All these aspects constitute a very grim scenario, mandating immediate action to reverse it.

Ice thermal storage systems are receiving greater acceptance throughout the industry as an effective means of shifting summer peak power in commercial office buildings and providing lower operating costs [67]. Baseload power for freezing ice is taken off-peak at night. This also has a positive effect on power-generating plant efficiency, because of the higher load factor and constant operation, with minimum on and off daily cycles [68]. In addition, with a totally integrated off-peak ice storage design, it is possible to offset the relatively high kW/ton required to make ice by reducing pump and fan horsepower [69].

Off-peak air conditioning (OPAC) is already a major energy saver [68]. Partial storage with chiller priority is the most common application. Ice is produced only as a back-up which, in most climates, is only 10–15% of the time. Even during these times, the chiller operates during the day at 5–7% more efficiency than with no storage. This is because the chiller, being upstream of the ice storage, is handling the higher half of cooling the return brine temperature (i.e. 16 to 11°C). When the chiller is operating to build ice, it is at full constant load (no unloading, part loading or on/off operation).

Taking advantage of 1–3°C chilled water available with ice storage, equipment cost can be minimized throughout the water and air distribution systems. The installation of a full storage, dynamic ice harvester shifting a daily load of 300 tons, a 340 m^3 ice storage tank, a heat exchanger, and a low-temperature variable-air-volume (VAV) air distribution system in a 17,000 m^2 commercial office building [67] has proved the applicability of such systems. Substantial energy savings and utility costs result from reduced electrical demand and use of less expensive off-peak energy. Additional benefits include improved air quality levels, since low water temperatures at the cooling coil and low air-temperature distribution cause reduction in the growth of bacteria and fungus in the air ducts. Overall, advances in ice storage technology are expected to reduce the energy used for mechanical cooling of buildings by about 15% in the 1990s [68].

Despite these positive prospects, in recent years the A/C industry has been facing a major shake-up, possibly the worst in its history. Ratification of international agreements such as the 1987 Montreal Protocol on the phase out of the commonly used chlorofluorocarbon (CFC) refrigerants in A/C systems have created a major problem for the A/C and related industries as they try to identify and use new working fluids. Certain CFCs have been related to the depletion of the ozone layer and the greenhouse effect. Guidelines have been set for a steep reduction of fully halogenated CFCs, such as R-11, 12, 113, 114, 115 and halons 1211, 1301 and 2402.

The members of the European Union agreed in 1990 on an immediate 85% cut in the production of CFCs and the elimination of their use by the end of the century. In the USA, the Environmental Protection Agency has adopted a regulation on the Protection of Stratospheric Ozone, based on contributions from the industry and its representatives, various engineering societies, the Department of Energy and other interested parties. The freeze on fully halogenated CFCs took effect in the summer of 1989 by setting production to 1986 levels, while 20 and 30% reductions will follow over a ten-year period. Efforts are being made to shorten this time frame.

Consequently, other refrigerants with good thermophysical properties which satisfy the new environmental restrictions must be made available to replace those phased out. Identification and testing of substitute refrigerants has become a top priority to the air-conditioning and refrigeration industry. Partially halogenated refrigerants (hydrogenated chlorofluorocarbons), such as HCFC-22, are considered safer. Refrigerants not containing chlorine (hydrogenated fluorocarbons), such as HFC-152a and 134a are safest. Furthermore, the concept of employing a multicomponent refrigerant (mixtures) introduces a desirable degree of freedom in developing substitute refrigerants.

The selection of alternative working fluids for vapour compression processes depends on the particular application and on some distinctive properties, according to the chemical nature of the refrigerant, its thermophysical properties and on their compliance with health, safety and environmental restrictions. A comprehensive approach for the selection and screening of novel refrigerants is described in [70].

Three of the most promising candidates among the existing refrigerants for future use are HCFC-123, 124 and HFC-134a. These alternative refrigerants satisfy our environmental concerns, since they have about 98% less potential to deplete ozone than R-11 (commonly used in industrial refrigeration systems).

For an acceptable capacity of a vapour compression cycle, the cooling effect of the substitute refrigerants must not be much lower, and the specific volume must not be much higher, than the corresponding properties of the original equipment refrigerant. The COP cannot be much lower without also degrading the ability of the system to, at least temporarily, meet condenser and evaporator temperatures. The efficiency of the compressor should not be much degraded if an otherwise acceptable refrigerant is employed.

CURRENT TECHNOLOGY FOR NATURAL VENTILATION IN BUILDINGS

Natural ventilation is the predominant passive cooling technique. In general, air ventilation (passive, hybrid or mechanical) of indoor environments is also necessary in order to maintain the required levels of oxygen in the space and the maintenance of acceptable indoor air quality levels.

Traditionally, ventilation requirements were achieved by natural means. In the majority of older buildings, infiltration levels were such as to allow considerable amounts of outdoor air to enter the building, while additional requirements were satisfied by simply opening the windows.

Modern architecture and energy-conscious design of buildings has reduced to a minimum the air infiltration inside buildings, in an attempt to reduce its impact on the cooling or heating load. Better construction of buildings has resulted in actually sealing the building from the outdoor environment. In particular, the construction of large glass office buildings, where the opening of windows is not allowed, has practically eliminated the possibility of using natural ventilation techniques for supplying fresh air to indoor spaces.

It is clear that building design approaches have to be re-evaluated and adapted to the new requirements, in order to facilitate the application of natural ventilation techniques for cooling purposes. The following discussion addresses the principles of the most up-to-date techniques for natural cooling.

The use of natural ventilation, provided that outdoor climatological conditions are favourable, can provide for the following:

- a reduction of the cooling load (Chapter 8);
- enhancement of thermal comfort conditions (Chapter 6); and
- maintenance of proper indoor air quality levels [71, 72].

The effectiveness of natural ventilation techniques is determined by:

- prevailing outdoor conditions – microclimate (wind speed, temperature, humidity and surrounding topography), and
- the building itself (orientation, windows – number, size, location).

Current trends of natural ventilation

Current trends for the successful implementation of natural ventilation techniques for cooling purposes are presented in the following discussion. A detailed analysis of the various concepts involved along with the information needed by building designers for their proper application, are analysed in Chapter 8.

New buildings allow for 0.2–0.5 air changes per hour (ACH) by infiltration, while with the windows wide open during summer it is possible to achieve 15–20 ACH. Even larger air changes, around 30 ACH, can be achieved by natural means, but there is a need for a large number of window openings and careful placement within the space. The overall problem of energy conservation in a building becomes more complex since a large number of openings will increase the thermal load during winter, by increasing thermal losses, since a glass window can not compare in its thermal performance with a well insulated wall. It is also important to link the necessary number of windows and their placement with the requirements for natural lighting purposes.

Natural ventilation is used to create a volumetric flow, for renewal of indoor air quantities and transfer of heat, resulting from the outdoor wind speed and/or stack effects, in order to:

- remove heat stored into the building materials; and
- enhance heat dissipation from the human body to the environment.

The savings from natural ventilation depend on:

- the number of air changes;
- the construction of the building (heavy or light construction);
- the microclimate; and
- the temperature and humidity in the space.

Outdoor temperature, humidity and wind velocity are determinant factors for the successful application of natural ventilation techniques. For cooling purposes, the incoming air should be at a lower temperature than the indoor air temperature. However, even at higher temperatures, the resulting air flow inside the space can cause a positive effect on the thermal comfort conditions of the occupants, since it increases heat dissipation from the human body and enhances evaporative and convective heat losses. An increase of air velocity by 0.15 m s^{-1} compensates an increase of 1°C at a relative humidity of 75%. There is a direct link between thermal comfort and ventilation rates, which is explained in detail in Chapter 6.

Natural ventilation techniques for cooling purposes are also very effective during the night hours, when outdoor air temperatures are usually lower than the indoor ones. As a result the cooling load is reduced and the peak indoor air temperatures can be reduced by 1 to 3°C.

Air humidity is the most important limiting factor for the application of natural ventilation techniques. High levels of humidity have a negative influence on thermal comfort. As a result, in regions with high relative humidity levels during summer, the use of conventional air-conditioning systems is necessary in order to remove water vapour from indoor air (dehumidification). Under such circumstances, natural ventilation during the day- or night-time hours should be avoided.

Overall, natural ventilation research has concentrated on the design and optimization of window openings, for enhancing natural ventilation in relation to daylighting, solar gains, and thermal losses during winter. Dealing with the prediction of the wind potential in urban environments and air movement within the various spaces inside a building are areas which require additional research. The European Commission research programme for passive cooling of buildings, PASCOOL [73], has addressed most of these problems and reported significant advances.

In addition to the use of windows and other openings for natural ventilation, there are some additional means of enhancing the air movement. Wind towers utilize the kinetic energy of wind, which is properly channelled within the building in order to generate air movement within a space. They have been successfully integrated in many architectural designs [74, 75]. Solar chimneys are constructions used to promote air movement throughout the building using solar gains. They are positioned on the sunward side of the building to make the best possible use of direct solar gains. Performance information is provided in [76, 77].

To increase the cooling effectiveness of natural ventilation techniques, especially at locations with low outdoor air velocity and variable wind directions, it is possible to incorporate wing-walls into the building's design [78]. Wing-walls simply project outward next to a window and even a slight breeze against the wall creates a high-pressure zone on one side and a low one on the other. The pressure differential draws the outdoor air in through one open window and out through the adjacent one. This technique is most suitable for providing increased ventilation rates in rooms with windows on only one side, as is the case in most schools. Classrooms are usually next to long hall-ways allowing windows on one side only, which reduces the possibility for cross ventilation. As a result, a number of schools in southern USA have been built with wing-walls, with very satisfactory results. Especially in Florida, where the law requires that new schools be designed with a non-mechanical method of venting classrooms, this concept has received wide acceptance.

Ceiling fans can also be used to enhance indoor thermal comfort conditions, by controlling indoor air circulation. Ceiling fans allow for higher indoor air temperatures and result in reduced cooling requirements by enhancing comfort conditions. They have been proved to be a viable technology and are especially attractive because of their low capital, operational and maintenance costs.

For a detailed overview of current practice in the use of natural ventilation in passive cooling, the reader should also refer to [9, 10, 79] and Chapter 9.

CONCLUSIONS

Natural and passive cooling techniques in buildings have been proved successful through numerous demonstration and pilot projects. They exhibit a high potential for reducing the energy consumption for cooling in commercial and residential buildings, which represents a significant percentage of the total energy consumption in buildings.

Technology advances have provided air-conditioning systems with high efficiency, good overall performance and lower initial cost. However, the current concerns about reducing energy consumption, the increased cost of energy and inherent problems of mechanical compression systems, which operate with refrigerants that have been linked to atmospheric pollution, have caused a re-examination of alternative technologies and systems. Natural and passive cooling techniques, which have been applied successfully in the past, are being revitalized, enhanced with new research findings and applied with current technology and advanced system-design know-how.

Passive cooling refers to techniques which can be used to prevent and modulate heat gains. Protection from heat gains involves the use of microclimate and site design, solar control, building form and layout, thermal insulation, behavioural and occupancy patterns and internal gain control. Heat-gain modulation can be achieved by properly using the thermal mass of the building itself (thermal inertia) in order to absorb and store heat during the daytime hours and return it to the space at a later time.

Natural cooling refers to the use of natural heat sinks for excess heat dissipation from interior spaces, including ventilation, ground cooling, evaporative cooling and radiative cooling.

Combining different passive and natural cooling techniques, it is possible to prevent overheating problems, decrease cooling load and improve indoor thermal comfort conditions.

In particular, it is possible to achieve an overall 20% energy conservation in new and existing buildings. Significant energy savings for cooling can be achieved, for example by reducing external loads using proper shading of the building facade, by reducing internal loads from lighting with the use of energy-efficient fluorescent lamps, by using natural cooling techniques like ground, evaporative and radiative cooling and by night ventilation.

Substantial energy savings can also be achieved in existing mechanical cooling systems. Proper maintenance of air-conditioners can reduce electrical peak demand, save energy and consequently reduce operating costs. Individual repair measures may include, for example, correcting low air flow rates, repairing overcharging and undercharging of refrigerant and repairing duct leakage. Heat pumps and other high performance mechanical systems can replace older technology equipment in order to achieve considerable energy savings for applications where air-conditioning systems are necessary.

REFERENCES

1 McQuiston, F.C. and J.D. Parker (1982). *Heating, Ventilating and Air Conditioning. Analysis and Design*, 2nd edition, John Wiley & Sons, New York.

2 Threlkeld, J.L. (1970). *Thermal Environmental Engineering*, 2nd edition, Prentice Hall, Englewood Cliffs, NJ.

3 Fischer, S. (1991).'Energy use impact of CFC alternatives', *Energy Engineering*, Vol. 88, No. 3, pp. 6–21.

4 Van Wylen, G.J. and R.E. Sonntag (1978). *Fundamentals of Classical Thermodynamics*, 2nd edition, John Wiley & Sons, New York.

5 Sauer, H. and R.H. Howell (1983). *Heat Pump Systems*, John Wiley & Sons, New York.

6 Santamouris, M. (ed.) (1990). 'Horizontal study on passive cooling', *CEC-Building 2000 Project*, Chapter 1, pp. 1–7, organized by the Directorate General 12, Commission of the European Communities.

7 Goulding, J.R., J. Owen Lewis and T.C. Steemers (eds) (1992). *Energy in Architecture, The European Passive Solar Handbook*, Commission of the European Communities, Brussels.

8 Cook, J. (1990). 'The state of passive cooling research', *Passive Cooling*, ed. J. Cook, Chapter 7, pp. 539–569. MIT Press, Cambridge, MA.

9 Antinucci, M., B. Fleury, J. Lopez d'Asiain, E. Maldonado, M. Santamouris, A. Tombazis and S. Yannas (1992).'Passive and hybrid cooling of buildings. State of the art', *International Journal of Solar Energy*, Vol. 11, pp. 251–272.

10 Steemers, T.C. (1991). 'The state of the art in passive cooling', *International Journal of Solar Energy*, Vol. 10, pp. 5–14.

11 Moffat, A. and M. Schiller (1981). *Landscape Design. How to Save Energy.* William and Narrow Company, New York,

12 Yannas, S. (1990). 'Solar control techniques', *Proc. Workshop on Passive Cooling*, eds E. Aranovich, E. Oliveira Fernandes and T.C. Steemers, Ispra, Italy, 2–4 April, pp. 75–97.

13 Markus, T.A. and E.N. Morris (1980). *Building, Climate and Energy.* Pitman, London.

14 Antinucci, M. (1990). 'Heat attenuation as a means to limit cooling load and improve comfort', *Workshop on Passive Cooling*, eds E. Aranovitch, E. Fernades and T. Steemers, Ispra, Italy, pp. 111–119.

15 Braun, J. (1990). 'Reducing energy costs and peak electrical demand through pptimal control of building thermal storage', *ASHRAE Transactions*, Vol. 96 No. 2, pp. 876–888.

16 Brown, M. (1990). 'The thermal mass of buildings in reducing energy consumption', *ASME Journal of Solar Energy Engineering*, Vol. 112, p. 273.

17 Chandra, S. (1990). 'Ventilative cooling', *Passive Cooling*, ed. J. Cook, Chapter 2, pp. 42–84. MIT Press, Cambridge, MA.

18 Chandra, S., P. Fairey and M. Houston, (1986). 'Cooling with ventilation', Florida Solar Energy Center, SERI/SP-273-2966, DE86010701.

19 Labs, K. (1979). 'Underground building climate', *Solar Age*, Vol. 4, pp. 44–50.

20 Labs, K. (1989). 'Earth coupling', *Passive Cooling*, ed. J. Cook, Chapter 5, pp. 197–346. MIT Press, Cambridge, MA.

21 Fanchiotti, A. and G. Scudo (1981). 'Large scale underground cooling system in Italian 16th ventury Palladian villa', American Solar Energy Society Passive Cooling Conference, Miami.

22 Tzaferis, A., D. Liparakis, M. Santamouris and A. Argiriou, (1992). 'Analysis of the accuracy and sensitivity of eight models to predict the performance of earth-to-air heat exchangers', *Energy and Buildings*, Vol. 18, pp. 35–43.

23 Givoni, B. and L. Kats, (1985). 'Earth temperature and underground buildings', *Energy and Buildings*, Vol 8, pp. 15–25.

24 Yellott, J.I. (1989). 'Evaporative cooling', *Passive Cooling*, ed. J. Cook, Chapter 3, pp. 85–137. MIT Press, Cambridge, MA.

25 Velazquez, R., J. Guerra, S. Alvarez and J.M. Cejudo (1991). 'Case Study of Outdoor Climatic Comfort: The Palenque at EXPO'92', PLEA'91, Seville, Spain.

26 Guerra, J., J.L. Molina and J.M. Cejudo (1991). 'Thermal performance of water ponds: modelling and cooling applications', PLEA'91, Seville, Spain.

27 Alvarez, S., E.A. Rodriguez and J.L. Molina (1991). 'The Avenue of Europe at EXPO'92: Application of cool towers in architecture and urban spaces', PLEA'91, Seville, Spain.

28 Rodriguez, E., S. Alvarez and R. Martin, (1991). 'Direct air cooling from water drop evaporation', PLEA'91, Seville, Spain.

29 Rodriguez, E.A., S. Alvarez and R. Martin, (1991). 'Water drops as a natural cooling resource. physical principles', PLEA'91, Seville, Spain.

30 Bahadori, M.N. (1986). 'Natural air-conditioning systems', *Advances in Solar Energy*, Vol. 3, pp. 283–356, Plenum Press, New York.

31 Cook, J. (1989). 'Phoenix Solar Oasis: cooling design of urban spaces', *14th National Passive Solar Conference*, Denver, pp. 350–355.

32 Kent K. and T. Lewis (1990). 'Natural Draft Evaporative Cooling', *15th National Passive Solar Conference*, pp. 157–160, Austin.

33 Alvarez, S., E.A. Rodriguez and J.L. Molina (1991). 'The Avenue of Europe at EXPO'92: Application of cool towers', PLEA'91, Seville, Spain.

34 Martin, M. (1989). 'Radiative cooling', *Passive Cooling*, ed. J. Cook, Chapter 4, pp. 138–196. MIT Press, Cambridge, MA.

35 McCathren, J.R. and J.M. Akridge, (1982). 'Radiant cooling to the night sky in hot, humid climates', *Progress in Passive Solar Energy Systems*, American Solar Energy Society, pp. 877-882,

36 Santamouris, M. (1990). 'Natural cooling techniques', *Proceedings of Passive Cooling Workshop*, eds E. Aranovitch, E. de Oliveira Fernades, T.C. Steemers, Ispra, pp. 143–153.

37 Givoni, B. (1981). *Proceedings of the. International Passive Hybrid Cooling Conference*, eds A. Bowen, E. Clark and K. Labs, Miami Beach, p. 279.

38 Hay, H.R. and J.I. Yellott (1969). 'Natural air conditioning with roof ponds and movable insulation', *ASHRAE Transactions*, Vol. 75, No. 1, pp. 165–177.

39 Argiriou, A., M. Santamouris, C.A. Balaras and S.M. Jeter (1993). 'Potential of radiative cooling in southern Europe', *International Journal of Solar Energy*, Vol. 13, pp. 189–203.

40 Catalanotti, S., V. Cuomo, G. Piro, D. Ruggi, V. Silvestrini and G. Troise (1975). 'The radiative cooling of selective surfaces', *Solar Energy*, Vol. 17, pp. 83–89.

41 Bartoli, B., B. Catalanotti, B. Coluzzi, V. Cuomo, V. Silvestrini and G. Troise, (1977). 'Noctural and diurnal performances of selective radiators', *Applied Energy*, Vol. 3, pp. 267–286.

42 Papadakis, G., G. Voulgaris, A. Frangoudakis and S. Kyritsis (1988). 'Night sky radiation in Athens during the summer. influence of city pollutants', *International Journal of Solar Energy*, Vol. 6, pp. 279–289.

43 Agas, G., T. Matsaggos, M. Santamouris and A. Argiriou (1991). 'On the use of the atmospheric heat sinks for heat dissipation', *Energy and Buildings*, Vol. 17, pp. 321–329.

44 *Energy in Europe* (1992). Directorate General for Energy, DG XVII, Commission of the European Communities, Brussels.

45 Santamouris, M., A. Argiriou, E. Dascalaki, C.A. Balaras and A. Gaglia, (1994). 'Energy characteristics and savings potential in office buildings', *Solar Energy*, Vol. 52, pp. 59–66.

46 *Passive Solar Energy as a Fuel* (1990). The Commission of the European Communities, Directorate General XII for Science Research and Development, Brussels.

47 Santamouris, M., A. Argiriou, E. Dascalaki, M. Vallindras, A. Gaglia and J. Sigalas (1992). 'Energy conservation in office buildings', Final Report, Greek Productivity Centre and Ministry of Industry, Research and Technology.

48 Santamouris, M., E. Dascalaki, C.A. Balaras, A. Argiriou and A. Gaglia (1993). 'Performance assessment and the potential for energy conservation and the use of alternative energy sources in buildings', *Proceedings of 3rd European Conference on Architecture, Solar Energy in Architecture and Urban Planning*, 17–21 May, Florence.

49 Santamouris, M., E. Dascalaki, C.A. Balaras, A. Argiriou and A. Gaglia (1994). 'Energy consumption and the potential for energy conservation in school buildings, in Greece', *Energy, The International Jourmal*, Vol. 19, pp. 653–660.

50 Santamouris, M., E. Dascalaki, C.A. Balaras, A. Argiriou and A. Gaglia (1994). 'Energy performance and energy conservation in health care buildings in Greece', *Energy Conversion and Management*, Vol. 35, pp. 293–305.

51 Campbell, J. (1988). 'Use of Passive Solar Energy in Offices', *Passive Solar Energy in Buildings*, ed. P. O'Sullivan. The Watt Committee on Energy, Elsevier Applied Science Publishers, New York.

52 Lefebvre, R.R. (1993). 'New HVAC system reduces operating costs', *ASHRAE Journal*, Vol. 35, No. 4 (April), pp. 20–23.
53 Boldt, J.G. (1993). 'Separate HVAC systems maximize energy efficiency', *ASHRAE Journal*, Vol. 35, No. 4 (April), pp. 16–19.
54 *Communes, Bilan Energetique du Patrimoine et des Services, Dépenses & Consommations 1990* (1990). Ministère de l'Interieur et de la Securité Publique, Agence de l'Environnement et de la Maîtrise de l'Energie.
55 Balaras, C.A. (1993). 'Energy Efficient Building in Greece', in *The European Directory of Energy-Efficient Building*, ed. B. Cross. James & James Science Publishers, London.
56 McNall, P., E. Pierce and J. Barnett, (1978). 'Control Strategies for Energy Conservation', in *Energy Conservation Strategies in Buildings*, ed. J.A. Stolwijk. J.B. Pierce Foundation.
57 Fleming, W.S. (1979). 'Energy conservation: an investigation of the thermal comfort alternative', *ASHRAE Transactions*, Vol. 85, No. 2, pp. 813–824.
58 Zmeureanu, R. and A. Doramajian (1992). 'Thermally acceptable temperature drifts can reduce the energy consumption for cooling in office buildings', *Building & Environment*, Vol. 27, No. 4, pp. 469–481.
59 Proctor, J. (1991). 'An ounce of prevention: residential cooling repairs', *Home Energy*, Vol. 8, No. 3, pp. 23–28.
60 Amberger, R.F. and J.A. DeFrees (1993). 'retrofit cogeneration system increases refrigeration capacity', *ASHRAE Journal*, Vol. 35, No. 4 (April), pp. 24–27.
61 Fiorino, D.P. (1993). 'Chilled water storage system reduces energy costs', *ASHRAE J.*, Vol. 35, No. 4 (April), pp. 30–33.
62 Meier, A. (1991). 'Variable-speed drive: Energy winner, power loser', *Home Energy*, Vol. 8, No. 3, pp. 8–9.
63 King, A. '(1993). The growing market for air conditioning', *Proceedings: Advanced Systems of Passive and Active Climatisation*, 3–5 June, Barcelona.
64 Balaras, C.A., M. Santamouris, M. Vafiadis, P. Vovolis, O. Voumdas, D. Ragousis (1991). 'Correlation of bioclimatic indices with peak electric load, in the Athens area for the period of 1986–1989', Institute for Technological Applications, Greek Productivity Centre, Athens.
65 United States Department of Energy (1985). *Non-residential Buildings Energy Conservation Survey: Characteristics of Commercial Buildings*, Washington, DC, pp. ix, 21, 105.
66 Al-Jamal, K., O. Alameddine, H. Al-Shami and N. Shaban (1988). 'Passive cooling evaluation of roof pond systems', *Solar & Wind Technology*, Vol. 5, pp. 55–65.
67 Landry, C.M. and C.D. Noble (1991). 'Making ice thermal storage first-cost competitive', *ASHRAE Journal*, Vol. 33, No. 5, pp. 19–22.
68 MacCracken, C.D. (1991). 'Off peak air-conditioning: A major energy saver', *ASHRAE Journal*, Vol. 33, No. 12, pp. 12–23.
69 Harmon, J.J. and H. Chung Yu (1991). 'Centrifugal chillers and glycol ice thermal storage units', *ASHRAE Journal*, Vol. 33, No. 12, pp. 25–31.
70 Balaras, C.A. and S.M. Jeter (1991). 'A methodology for selecting and screening novel refrigerants for use as alternative working fluids, *Energy Conversion and Management*, Vol. 31, No. 4, pp. 389–398.
71 Argiriou, A., C.A. Balaras, E. Dascalaki, A. Gaglia, G. Gountelas, K. Moustris, M. Santamouris and M. Vallindras, (1992). 'A survey of indoor air quality in office buildings in Athens, Greece', *Proceedings of the International Conference on Indoor Air Quality and Ventilation*, Athens.

72 Balaras, C.A., M. Santamouris, E. Dascalaki, A. Argiriou, D. Assimakopoulos, I. Tselepidaki and M. Loizidou (1993). 'indoor air quality and health symptoms in town halls in Athens, Hellas', *Proceedings of Indoor Air '93, 6th International Conference on Indoor Air Quality and Climate*, Helsinki, 4–8 July.

73 Santamouris, M. and A. Argiriou (1993). 'The CEC Project PASCOOL', *Proceedings of 3rd European Conference on Architecture, Solar Energy in Architecture and Urban Planning*, 17–21 May, Florence.

74 Bahadori, M. (1988). 'A passive cooling heating system for hot arid regions', *Proceedings of 13th National Passive Solar Conference*, Cambridge, MA, June, pp. 364–367.

75 Karakatsanis, C., M. Bahadori and B. Vickery (1986). 'Evaluation of pressure coefficients and estimation of air flow rates in buildings employing wind towers', *Solar Energy*, Vol. 37, pp. 363–374.

76 Bouchair, A. (1988). 'Moving air using stored solar energy', *Proceedings of 13th National Passive Solar Conference*, Cambridge, MA, June.

77 Dimoudi, A. (1989). 'An evaluation of the use of solar chimneys to promote the natural ventilation of buildings', M.Sc. Thesis, Cranfield Institute of Technology.

78 Melody, I. (1987). 'Solar cooling research', *Solar Today*, January, pp. 23–24.

79 Fleury, B. (1990). 'Ventilative cooling: State of the art', *Proceedings of Workshop on Passive Cooling*, Ispra, Italy, 2–4 April, pp. 123–133.

2

Passive cooling of buildings

Buildings, although static in space, are dynamically related to time. In addition to offering a shelter and fulfilling aesthetic criteria, they should ensure means of comfort (thermal, visual, acoustic) for their inhabitants.

The thermal behaviour of buildings is affected by various parameters. These include the climatological ones, which are environmental variables and are not subject to human control. The other type of parameters is the design variables, which are under control at the design stage. Insufficient attention to the aspect of a building's thermal behaviour at the first stages of its design can lead to an inhospitable internal environment. During summer, especially in climates with hot weather, buildings are exposed to high intensities of solar radiation and high temperatures. This may result in overheating conditions that exceed the threshold of thermal comfort in the interior of buildings. Under these conditions, cooling of buildings is of great importance.

Traditional architecture can show examples of harmonization with local climate. In hot and dry climates the buildings were of massive construction with few and small openings and light colours on the external surface. Fountains, pools, water streams and vegetation accomplished the cooling effect through evaporation. In hot and humid climates, where ventilation is desirable, we find lightweight structures with large openings and large overhangs. Modern buildings, in many cases, fail to follow the examples set by tradition. Part of the blame for the failure, from the climatic point of view, has been attributed to the so called 'international style', that brought science and technology to its design, adapting design ideas and features regardless of different climatic regions. This was connected with the separation of the envelope's design, which was the task of the architect, and the interior operation, which was entirely left to service engineers. This approach led to a total dependence on mechanical equipment to support the energy needs of buildings. With the present state of the art of air conditioning, the cooling needs of any building can be met, but at the cost of using unnecessary amounts of energy.

BENEFITS OF PASSIVE COOLING

The energy required for heating and cooling of buildings is approximately 6.7% of the total world energy consumption. By proper environmental design, at least 2.35% of the world energy output can be saved [1]. In hot climate countries, energy needs for cooling can amount to two or three times those for heating, on an annual basis [2]. Utilization of the basic principles of heat transfer, coupled to the local climate, and exploitation of the physical properties of the construction materials, could make possible the control of the comfort conditions in the interior of buildings. Even in areas with average maximum ambient temperature around 31.7°C, comfortable conditions inside buildings can be achieved by means of proper building design [3] that frequently makes the use of air-conditioning units in dwellings unjustified. It is estimated [4] that an increase of 9 mtoe (million tons of oil equivalent) per annum in the total technical potential solar contribution (all potential solar usage is exploited) is possible in all the EU countries by the year 2010, compared to 1990, if passive cooling is applied in dwellings.

Passive cooling strategies in the design of buildings should be considered, since the extensive use of air-conditioning units is associated with the following problems:

Environmental

- Wide use of air-conditioning units has caused a shift in electrical energy consumption to the summer season and an increased peak electricity demand. Peak electric loads impose an additional strain on national grids, which can only be covered by development of extra new power plants. It is estimated [5] that in the USA the total electric peak load induced by air-conditioning units is about 38% of the non-coincident peak load. In addition to environmental implications, increased electrical peak loads result in an increase of the average cost of electricity to cover the construction of new power stations.
- Increased electrical energy production contributes to exploitation of the finite fossil fuels, to atmospheric pollution and to climatological changes. During the production process (fuel conversion), CO_2, which is one of the main causes of the greenhouse effect, is released. Coal-fired power plants burn and emit approximately 0.5 kg of carbon in the form of CO_2 for each kWh generated [6]. From the pre-industrial period to 1987, the ever-increasing use of fossil fuel and deforestation together have raised atmospheric CO_2 concentration to some 24% [7].
- Heat rejection during the production process (for electrical energy and air-conditioning units) and from the operation of air-conditioning units themselves increases the phenomenon of the 'urban heat island' (the climate modification due to urban development, which produces generally warmer air in cities than the surrounding countryside). Studies [8] show that summer heat islands are of average daily intensity of 3–5°C, resulting in discomfort and increased air-conditioning loads. It is estimated [9] that 5 to 10% of the urban American electrical demand is for additional air conditioning to compensate for heat islands.
- Ozone-layer depletion can be caused by CFCs and HFCs (the most common refrigerants of currently used air-conditioning units) from possible leakage